

# Sol coding benchmark

## What do we buy with more effort?

Six tasks. Sol 5.6, Sol 6 and Sol 6.1. Low, Medium, High, Xhigh, Max and Ultra. Two independent attempts per task, Standard speed.

|                                                                                          |                                                                                                             |                                                                                             |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <div>ATTEMPTS</div> <div>216</div> <div>3 models x 6 efforts x 6 tasks x 2 repeats</div> | <div>API-EQUIVALENT TOTAL</div> <div>≥ \$87.59</div> <div>Observed token price; ≥ marks a lower bound</div> | <div>ACCEPTED</div> <div>213/216</div> <div>Completion and full functional acceptance</div> |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

### Six stories in the evidence

**The cost of better tests:** on MVCC, Sol 6.1 Low tests detect 4/6 and 5/6 fixed defects; Medium detects 5/6 and 6/6 for 9.9% more task cost. High detects all six in both attempts at about \$0.203 per MVCC attempt.

**Which defects were anticipated:** the matrix preserves every MVCC repetition and distinguishes detection, missed defects, no recognized added tests and unavailable evaluation.

**Implementation efficiency:** on the fixed DAG workload, Sol 6.1 Low to Ultra costs 2.11x as much to generate. Execution time falls 32.3% and peak Python allocation falls 9.8%.

**Effort premium:** Sol 6.1 Max costs 2.90x Low across the suite. Both score 100 and deliver 12/12 accepted completions; added tests remain a separate evidence layer.

**Repeat spending:** 105/106 exact-cost pairs have identical main grades, but 20 cost more than 1.5x as much in one repetition.

**Tariff and token usage:** Sol 6.1 costs 28.9% less than Sol 6 on 70 matched attempts per model. Repricing both usage traces at the Sol 6.1 schedule leaves a 16.0% difference.

Functional context: 16/18 settings score 100. The report leads with specific test, implementation and spending differences, each with its own definition. Manual code-design review remains pending.

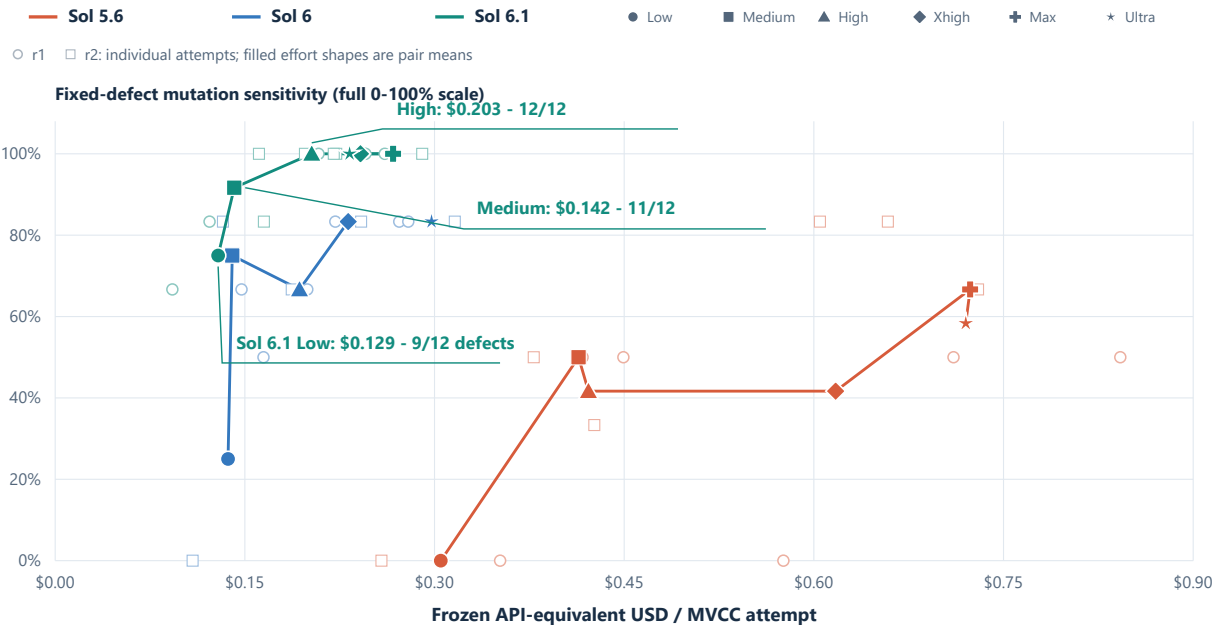
Money is a frozen Standard API token equivalent for observed Codex usage. It is not an invoice or a current-price quotation.

# The cost of better tests

MVCC is the leading case: all 36 saved implementations pass main acceptance, while their generated tests differ in detecting six fixed defects. X is task USD; Y is generated-test mutation sensitivity.

## The cost of tests that detect MVCC defects

One complex task, six fixed injected defects, two independent attempts per setting



Sol 6 Max: N/A as a pair (1/2 usable). The r1 observation remains visible; no mean is imputed.

**Sol 6.1 Low -> Medium: 9.9% more money; 9 -> 11 of 12 fixed-defect checks detected.**

Upper-left is preferable for this measured testing criterion. Lines follow effort order.

These 36 observations assess candidate-written tests on MVCC, not overall intelligence or code quality.

## Sol 6.1 progression

| Effort | USD / MVCC attempt | Defects: r1 / r2 | Pair sensitivity |
|--------|--------------------|------------------|------------------|
| low    | \$0.129            | 4/6 / 5/6        | 75.0%            |
| medium | \$0.142            | 5/6 / 6/6        | 91.7%            |
| high   | \$0.203            | 6/6 / 6/6        | 100.0%           |
| xhigh  | \$0.241            | 6/6 / 6/6        | 100.0%           |
| max    | \$0.267            | 6/6 / 6/6        | 100.0%           |
| ultra  | \$0.233            | 6/6 / 6/6        | 100.0%           |

Mean points require two available scores; both actual repetitions are also visible. Sol 6 Max has one unavailable evaluation and no pair mean. No added tests is zero for this criterion alone.

A 100% score means detecting these six seeded defects. This task-specific measure does not combine overall code design, general intelligence or production reliability.

# Which bugs did the model anticipate?

Each row is one MVCC candidate: 3 models x 6 efforts x 2 repetitions. The six columns identify the same controlled defects, preserving outcomes hidden by a pair average.

## Which MVCC defects did the tests detect?

Every attempt is shown; each column is one frozen contract mutation

| yes                      | Detected | no    | Missed         | none       | No added tests | N/A            | Unavailable |                     |
|--------------------------|----------|-------|----------------|------------|----------------|----------------|-------------|---------------------|
| Model / effort / attempt |          | Found | Snapshot reads | Write skew | Phantom reads  | Rollback reads | Recovery    | Checkpoint aliasing |
| Sol 5.6 Low - r1         | 0/6      | none  | none           | none       | none           | none           | none        | none                |
| Sol 5.6 Low - r2         | 0/6      | none  | none           | none       | none           | none           | none        | none                |
| Sol 5.6 Medium - r1      | 3/6      | no    | yes            | yes        | yes            | no             | no          | no                  |
| Sol 5.6 Medium - r2      | 3/6      | no    | yes            | yes        | yes            | no             | no          | no                  |
| Sol 5.6 High - r1        | 3/6      | no    | yes            | yes        | no             | yes            | no          | no                  |
| Sol 5.6 High - r2        | 2/6      | no    | yes            | yes        | no             | no             | no          | no                  |
| Sol 5.6 Xhigh - r1       | 0/6      | none  | none           | none       | none           | none           | none        | none                |
| Sol 5.6 Xhigh - r2       | 5/6      | yes   | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 5.6 Max - r1         | 3/6      | no    | yes            | yes        | no             | yes            | no          | no                  |
| Sol 5.6 Max - r2         | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 5.6 Ultra - r1       | 3/6      | no    | yes            | yes        | no             | yes            | no          | no                  |
| Sol 5.6 Ultra - r2       | 4/6      | no    | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 6 Low - r1           | 3/6      | no    | yes            | yes        | yes            | no             | no          | no                  |
| Sol 6 Low - r2           | 0/6      | none  | none           | none       | none           | none           | none        | none                |
| Sol 6 Medium - r1        | 4/6      | no    | yes            | yes        | yes            | no             | yes         | yes                 |
| Sol 6 Medium - r2        | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6 High - r1          | 4/6      | no    | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 6 High - r2          | 4/6      | no    | yes            | yes        | no             | yes            | yes         | yes                 |
| Sol 6 Xhigh - r1         | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6 Xhigh - r2         | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6 Max - r1           | 5/6      | yes   | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 6 Max - r2           | N/A      | N/A   | N/A            | N/A        | N/A            | N/A            | N/A         | N/A                 |
| Sol 6 Ultra - r1         | 5/6      | yes   | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 6 Ultra - r2         | 5/6      | yes   | yes            | yes        | yes            | yes            | no          | no                  |
| Sol 6.1 Low - r1         | 4/6      | no    | yes            | yes        | yes            | no             | yes         | yes                 |
| Sol 6.1 Low - r2         | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Medium - r1      | 5/6      | no    | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Medium - r2      | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 High - r1        | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 High - r2        | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Xhigh - r1       | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Xhigh - r2       | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Max - r1         | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Max - r2         | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Ultra - r1       | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |
| Sol 6.1 Ultra - r2       | 6/6      | yes   | yes            | yes        | yes            | yes            | yes         | yes                 |

Detected: candidate tests pass the implementation and positive control, then fail a mutant by assertion.

No added tests: numerical zero sensitivity. Unavailable: no numerical score; it is never replaced by zero.

Six fixed defects are a limited testing sample. Two attempts per setting show observations, not confidence intervals.

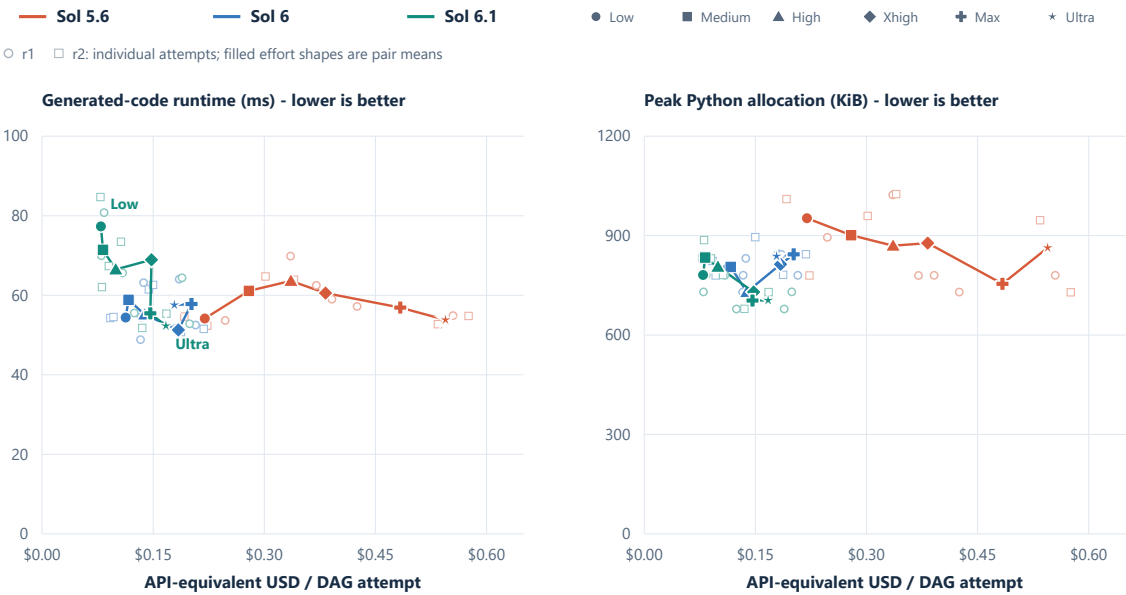
Detection requires an assertion failure against a mutant after candidate and positive-reference tests pass without skips. Errors/hangs are inconclusive. No added tests is explicit zero; unavailable is missing evidence.

# Same correctness, different efficiency

The DAG case separates the cost of generation from the runtime and allocation of saved code. All candidates pass main correctness. Lower runtime and allocation are better for this fixed workload.

## Correct DAG code can have different execution efficiency

Fixed workload: 1,500-task dependency chain; three worker limit



**Sol 6.1 Low -> Ultra: 2.11x generation cost; all four main grades are 100/100.**

**Runtime: 77.3 -> 52.3 ms. Peak Python allocation: 781.3 -> 704.7 KiB.**

Runtime uses the median of three timings per candidate, then the mean of two candidate medians.

Memory is tracemalloc peak Python allocation, not total process memory. One workload on one machine.

Both repeated candidates support the runtime direction; no general performance ranking follows from this task.

## Sol 6.1 on the 1,500-task DAG workload

| Effort | Generation USD | Code runtime, ms | Peak Python, KiB |
|--------|----------------|------------------|------------------|
| low    | \$0.079        | 77.3             | 781.3            |
| medium | \$0.082        | 71.4             | 833.8            |
| high   | \$0.099        | 66.5             | 805.9            |
| xhigh  | \$0.148        | 68.9             | 729.8            |
| max    | \$0.146        | 55.5             | 704.1            |
| ultra  | \$0.167        | 52.3             | 704.7            |

Each candidate runtime is the median of three sequential archived samples after one warm-up; each setting averages its two candidate medians. Allocation averages two separate tracemalloc peaks and excludes native memory.

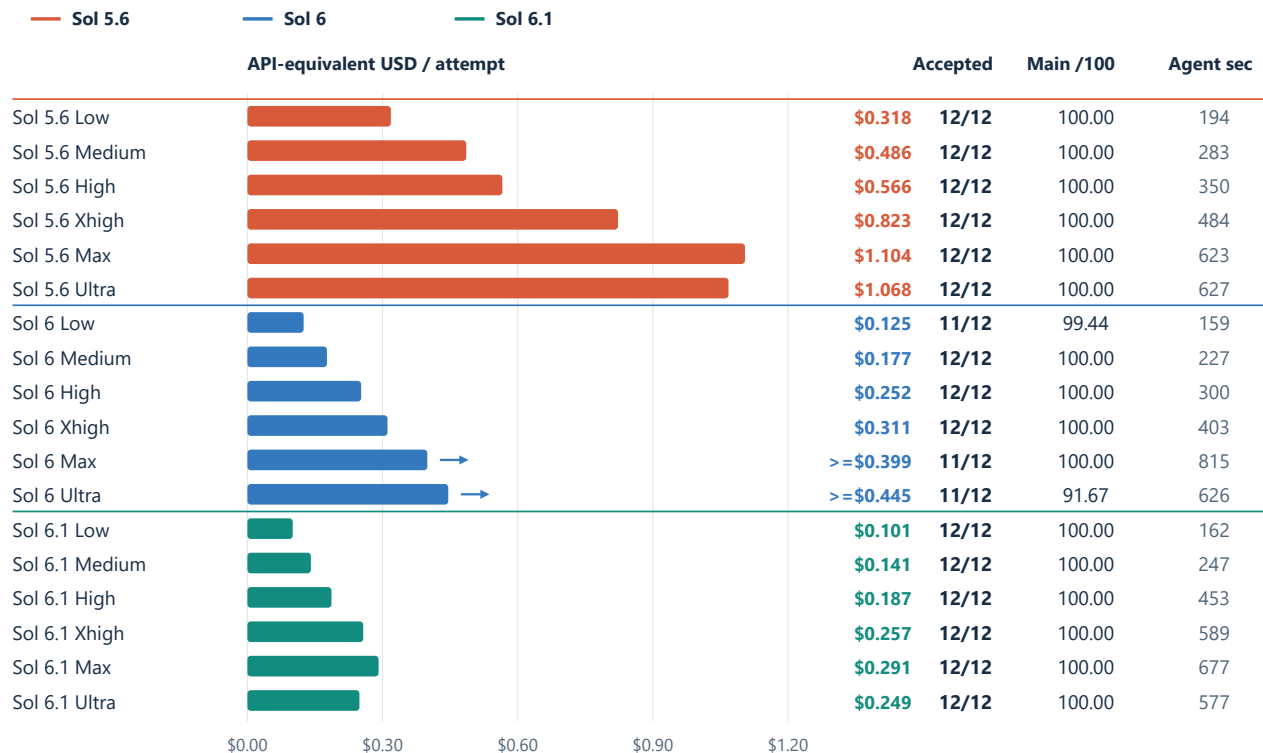
Both Sol 6.1 repetitions are faster at Ultra than Low on this workload. Performance profiles cover the original four tasks only; comparable optimizer/MVCC profiles were not collected.

# What does more effort buy?

Costs include all twelve attempts in each setting. Main functional grades and completed acceptance are shown beside money. Test sensitivity is a separate evidence layer.

## What does more effort buy across the six tasks?

Cost first; delivery, main functional score, and agent generation time kept as separate measures



**Sol 6.1 Max costs 2.90x Low; both have 12/12 accepted deliveries and 100/100 main score.**

The MVCC and DAG figures examine additional measured properties when the main functional score is at its ceiling.  
>=\$ and rightward arrows denote observed lower bounds. Agent seconds measure generation time, not code runtime.

## Sol 6.1 Low and Max: full-suite comparison

| Effort | USD / attempt | Main / 100 | Accepted | Agent seconds |
|--------|---------------|------------|----------|---------------|
| low    | \$0.101       | 100        | 12/12    | 161.7         |
| max    | \$0.291       | 100        | 12/12    | 676.8         |

Low to Max is a 2.90x observed cost premium and 4.19x mean agent-time multiple. Equal main grades leave room for differences in added tests, runtime and design. Higher effort does not guarantee a higher measured score.

Sol 6 Max and Ultra costs are incomplete, so bars show lower bounds. Sol 6.1 Ultra costs 14.59% less than Max in this archive; observed cost is not monotonically increasing with effort.

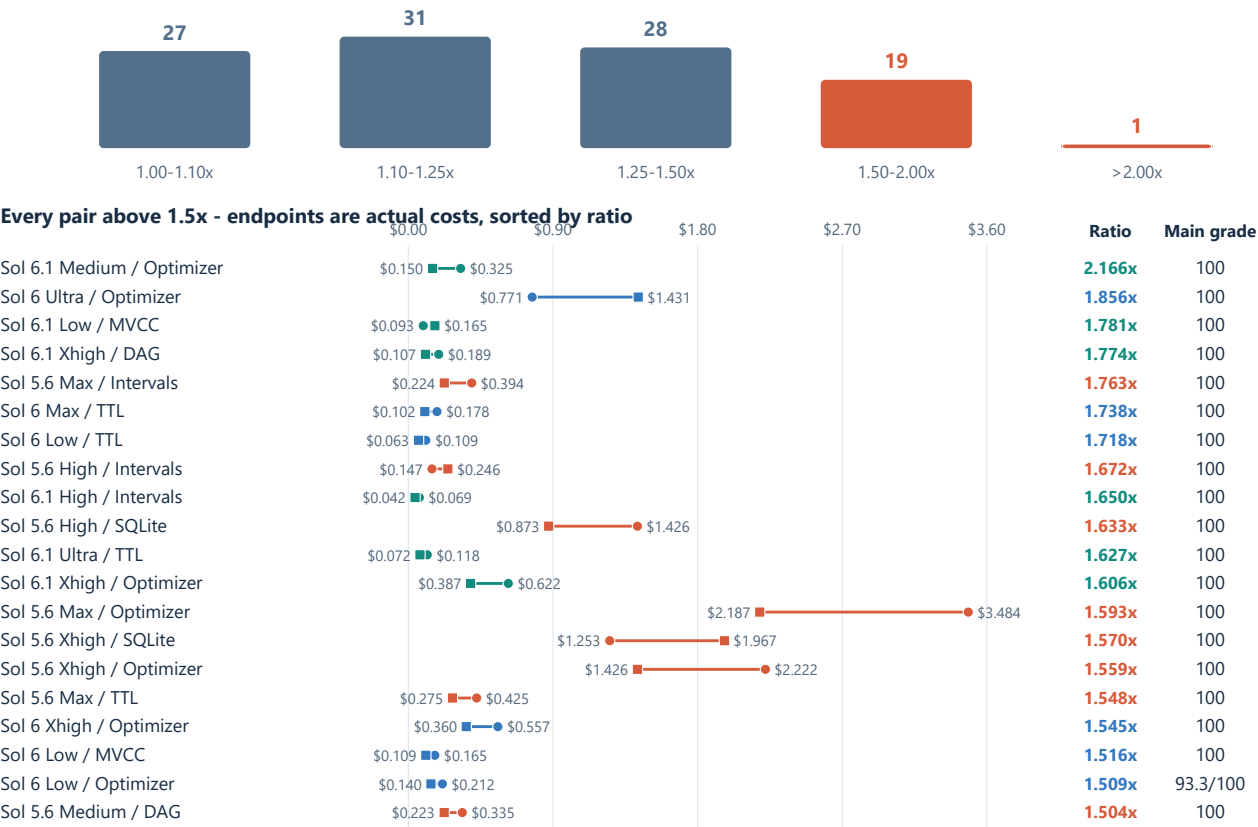
# Same answer, different spending

Of 108 repetition pairs, 106 have two exact USD estimates and 105 of those have identical main grades. The chart shows observed pairs rather than estimated confidence intervals.

## The same functional score can cost different amounts

105/106 complete-cost pairs have identical main grades; 20 cost ratios exceed 1.5x

All complete-cost pairs: larger / smaller actual repetition cost



**Two pairs have incomplete cost, so no exact spending ratio is calculated:**  
Sol 6 Max / SQLite - recorded observations remain in the archive.  
Sol 6 Ultra / Intervals - recorded observations remain in the archive.  
Circles: r1. Squares: r2. Two observations describe this sample; no confidence intervals or forecast are claimed.

Largest exact-cost ratio: Sol 6.1 / optimizer / Medium, \$0.150302 versus \$0.325499, or 2.166x. Both deliveries were accepted with a main score of 100.

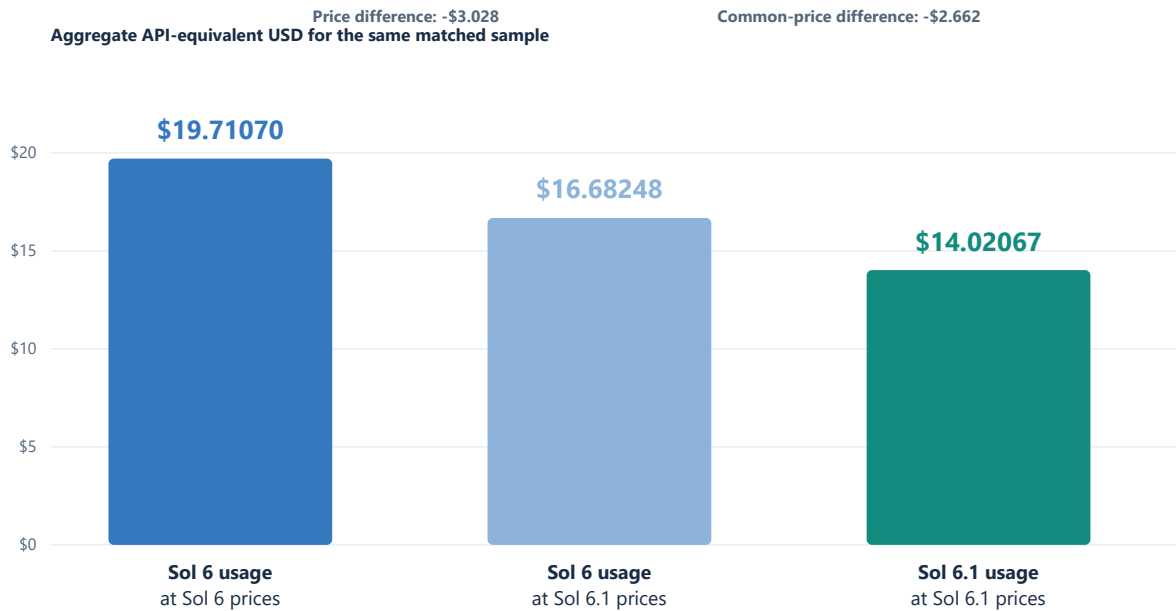
The two partial-cost pairs remain disclosed and archived. Token use, tools, recovery and service behavior can vary between runs. Two repetitions do not establish a long-run reliability or spending distribution.

# Tariff or token spending?

The same 70 task/effort/repetition keys are compared for Sol 6 and Sol 6.1. The bridge first reprices Sol 6 usage at the frozen Sol 6.1 schedule, then compares observed usage at common prices.

## Cheaper pricing, different token spending, or both?

70 matched task / effort / repetition observations - frozen API-equivalent USD



**Observed combined reduction: 28.9% (\$5.690) for these 70 matched observations.**

At common Sol 6.1 prices, recorded Sol 6.1 usage costs 16.0% less. The tariff step saves \$3.028.

Counterfactual repricing changes the tariff only. It does not simulate how a model would behave at another price. Cost estimates include recorded input, cached-input and output tokens. They are not actual Codex billing charges.

| Scenario                          | USD total / 70 attempts |
|-----------------------------------|-------------------------|
| Sol 6 usage at Sol 6 rates        | \$19.710704             |
| Same Sol 6 usage at Sol 6.1 rates | \$16.682480             |
| Sol 6.1 usage at Sol 6.1 rates    | \$14.020665             |

In this order of repricing, the tariff step reduces USD by \$3.028224, and the usage step by \$2.661815. Frozen cached-input rates are \$0.20 versus \$0.10 per million; other rates match.

Two incomplete Sol 6 observations and their matching Sol 6.1 keys are excluded from this view only. All 216 attempts remain archived. This order-dependent counterfactual does not establish causal attribution or intelligence.

# Dollars, quality and delivery

Twelve attempts per setting. USD / accepted includes spending on unsuccessful attempts.

| Model / effort   | USD / attempt | Functional / 100 | Accepted | USD / accepted | Mean time, s |
|------------------|---------------|------------------|----------|----------------|--------------|
| Sol 5.6 / low    | \$0.318       | 100.00           | 12/12    | \$0.318        | 194.2        |
| Sol 5.6 / medium | \$0.486       | 100.00           | 12/12    | \$0.486        | 282.8        |
| Sol 5.6 / high   | \$0.566       | 100.00           | 12/12    | \$0.566        | 350.1        |
| Sol 5.6 / xhigh  | \$0.823       | 100.00           | 12/12    | \$0.823        | 483.7        |
| Sol 5.6 / max    | \$1.104       | 100.00           | 12/12    | \$1.104        | 622.6        |
| Sol 5.6 / ultra  | \$1.068       | 100.00           | 12/12    | \$1.068        | 626.8        |
| Sol 6 / low      | \$0.125       | 99.44            | 11/12    | \$0.136        | 158.7        |
| Sol 6 / medium   | \$0.177       | 100.00           | 12/12    | \$0.177        | 226.7        |
| Sol 6 / high     | \$0.252       | 100.00           | 12/12    | \$0.252        | 300.2        |
| Sol 6 / xhigh    | \$0.311       | 100.00           | 12/12    | \$0.311        | 402.6        |
| Sol 6 / max      | ≥ \$0.399*    | 100.00           | 11/12    | >= \$0.435*    | 814.6        |
| Sol 6 / ultra    | ≥ \$0.445*    | 91.67            | 11/12    | >= \$0.486*    | 626.4        |
| Sol 6.1 / low    | \$0.101       | 100.00           | 12/12    | \$0.101        | 161.7        |
| Sol 6.1 / medium | \$0.141       | 100.00           | 12/12    | \$0.141        | 246.7        |
| Sol 6.1 / high   | \$0.187       | 100.00           | 12/12    | \$0.187        | 453.1        |
| Sol 6.1 / xhigh  | \$0.257       | 100.00           | 12/12    | \$0.257        | 588.6        |
| Sol 6.1 / max    | \$0.291       | 100.00           | 12/12    | \$0.291        | 676.8        |
| Sol 6.1 / ultra  | \$0.249       | 100.00           | 12/12    | \$0.249        | 577.1        |

Sol 6 / Max has a 40-minute reservation-service timeout; its saved code passes the functional and expanded checks. Sol 6 / Ultra has a provider capacity failure on the interval task. Sol 6 / Low completes every attempt but misses one optimizer scale check. These are distinct outcomes.

Displayed >= costs are observed lower bounds, rounded down. Accepted-cost totals and exact token estimates remain in the public data; incomplete final spending is not assigned zero.

# Task-level dollars and quality

Each cell shows mean API USD above mean functional score / 100, across two repetitions.

| Model / effort   | Intervals                 | Cache                   | DAG                     | Service                    | Optimizer               | MVCC                    |
|------------------|---------------------------|-------------------------|-------------------------|----------------------------|-------------------------|-------------------------|
| Sol 5.6 / low    | \$0.160<br>100.00 / 100   | \$0.210<br>100.00 / 100 | \$0.220<br>100.00 / 100 | \$0.554<br>100.00 / 100    | \$0.462<br>100.00 / 100 | \$0.305<br>100.00 / 100 |
| Sol 5.6 / medium | \$0.200<br>100.00 / 100   | \$0.250<br>100.00 / 100 | \$0.279<br>100.00 / 100 | \$0.856<br>100.00 / 100    | \$0.915<br>100.00 / 100 | \$0.414<br>100.00 / 100 |
| Sol 5.6 / high   | \$0.196<br>100.00 / 100   | \$0.328<br>100.00 / 100 | \$0.336<br>100.00 / 100 | \$1.149<br>100.00 / 100    | \$0.964<br>100.00 / 100 | \$0.422<br>100.00 / 100 |
| Sol 5.6 / xhigh  | \$0.259<br>100.00 / 100   | \$0.243<br>100.00 / 100 | \$0.383<br>100.00 / 100 | \$1.610<br>100.00 / 100    | \$1.824<br>100.00 / 100 | \$0.617<br>100.00 / 100 |
| Sol 5.6 / max    | \$0.309<br>100.00 / 100   | \$0.350<br>100.00 / 100 | \$0.484<br>100.00 / 100 | \$1.925<br>100.00 / 100    | \$2.836<br>100.00 / 100 | \$0.724<br>100.00 / 100 |
| Sol 5.6 / ultra  | \$0.244<br>100.00 / 100   | \$0.362<br>100.00 / 100 | \$0.545<br>100.00 / 100 | \$2.021<br>100.00 / 100    | \$2.515<br>100.00 / 100 | \$0.720<br>100.00 / 100 |
| Sol 6 / low      | \$0.046<br>100.00 / 100   | \$0.086<br>100.00 / 100 | \$0.113<br>100.00 / 100 | \$0.191<br>100.00 / 100    | \$0.176<br>96.67 / 100  | \$0.137<br>100.00 / 100 |
| Sol 6 / medium   | \$0.078<br>100.00 / 100   | \$0.118<br>100.00 / 100 | \$0.117<br>100.00 / 100 | \$0.310<br>100.00 / 100    | \$0.297<br>100.00 / 100 | \$0.140<br>100.00 / 100 |
| Sol 6 / high     | \$0.078<br>100.00 / 100   | \$0.098<br>100.00 / 100 | \$0.138<br>100.00 / 100 | \$0.515<br>100.00 / 100    | \$0.492<br>100.00 / 100 | \$0.193<br>100.00 / 100 |
| Sol 6 / xhigh    | \$0.102<br>100.00 / 100   | \$0.140<br>100.00 / 100 | \$0.184<br>100.00 / 100 | \$0.750<br>100.00 / 100    | \$0.459<br>100.00 / 100 | \$0.232<br>100.00 / 100 |
| Sol 6 / max      | \$0.157<br>100.00 / 100   | \$0.140<br>100.00 / 100 | \$0.202<br>100.00 / 100 | ≥ \$0.754*<br>100.00 / 100 | \$0.843<br>100.00 / 100 | \$0.300<br>100.00 / 100 |
| Sol 6 / ultra    | ≥ \$0.082*<br>50.00 / 100 | \$0.155<br>100.00 / 100 | \$0.179<br>100.00 / 100 | \$0.861<br>100.00 / 100    | \$1.101<br>100.00 / 100 | \$0.298<br>100.00 / 100 |
| Sol 6.1 / low    | \$0.050<br>100.00 / 100   | \$0.058<br>100.00 / 100 | \$0.079<br>100.00 / 100 | \$0.169<br>100.00 / 100    | \$0.118<br>100.00 / 100 | \$0.129<br>100.00 / 100 |
| Sol 6.1 / medium | \$0.066<br>100.00 / 100   | \$0.073<br>100.00 / 100 | \$0.082<br>100.00 / 100 | \$0.245<br>100.00 / 100    | \$0.238<br>100.00 / 100 | \$0.142<br>100.00 / 100 |
| Sol 6.1 / high   | \$0.055<br>100.00 / 100   | \$0.070<br>100.00 / 100 | \$0.099<br>100.00 / 100 | \$0.331<br>100.00 / 100    | \$0.361<br>100.00 / 100 | \$0.203<br>100.00 / 100 |
| Sol 6.1 / xhigh  | \$0.069<br>100.00 / 100   | \$0.100<br>100.00 / 100 | \$0.148<br>100.00 / 100 | \$0.480<br>100.00 / 100    | \$0.505<br>100.00 / 100 | \$0.241<br>100.00 / 100 |
| Sol 6.1 / max    | \$0.118<br>100.00 / 100   | \$0.117<br>100.00 / 100 | \$0.146<br>100.00 / 100 | \$0.509<br>100.00 / 100    | \$0.591<br>100.00 / 100 | \$0.267<br>100.00 / 100 |
| Sol 6.1 / ultra  | \$0.097<br>100.00 / 100   | \$0.095<br>100.00 / 100 | \$0.167<br>100.00 / 100 | \$0.404<br>100.00 / 100    | \$0.496<br>100.00 / 100 | \$0.233<br>100.00 / 100 |

Asterisks mark partial cost telemetry. These are saved-code functional grades; completed acceptance is reported separately. Task-level agent times appear in the secondary time comparison.

# Complex reasoning: behavioral evidence

Every small optimizer oracle and every MVCC check passed. Only one structured optimizer scale check failed.

| Model / effort   | Opt V | Opt C | Opt O | Opt S | MV Sem | MV Iso | MV Rec | MV V  |
|------------------|-------|-------|-------|-------|--------|--------|--------|-------|
| Sol 5.6 / low    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 5.6 / medium | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 5.6 / high   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 5.6 / xhigh  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 5.6 / max    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 5.6 / ultra  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / low      | 100.0 | 100.0 | 100.0 | 83.3  | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / medium   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / high     | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / xhigh    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / max      | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6 / ultra    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / low    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / medium | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / high   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / xhigh  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / max    | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |
| Sol 6.1 / ultra  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0  | 100.0  | 100.0  | 100.0 |

Optimizer: validation 20%, constraints 25%, optimality 35%, scale 20%. MVCC: semantics 25%, isolation 35%, recovery 25%, validation 15%. Values are category percentages; subcases fail their containing check.

The optimizer uses an independent exhaustive oracle on 64 seeded small problems per attempt, greedy traps, all tie rules and three structured scale fixtures. MVCC uses 480-step mixed transaction traces, 180-transaction replay traces, every checkpoint/log prefix, ABA, phantoms, write skew and failed-operation atomicity.

**Scale failure:** Sol 6 / Low, optimizer repetition 1, exceeded 8 seconds on the positive-bound fixture: 31 affordable projects and one individually infeasible high-value decoy. Its task score is 93.33; the paired task mean is 96.67. All its exact small-case answers passed.

A separate bounded replay reproduced the 8-second timeout. The positive control and Low repetition 2 completed the same fixture in about 0.20 and 0.22 seconds. The original grades were retained.

These tasks distinguish tested correctness and scaling behavior. Near-perfect outcomes across most settings leave too little separation to claim which model is generally smarter.

# Expanded correctness and found defects

The first four tasks retain the same quality-v2.1 checks and scores. Completed solutions pass 920/922 expanded checks.

| Model / effort   | Intervals | TTL/LRU cache | Concurrent DAG | SQLite service |
|------------------|-----------|---------------|----------------|----------------|
| Sol 5.6 / low    | 100.00    | 94.64         | 100.00         | 100.00         |
| Sol 5.6 / medium | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 5.6 / high   | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 5.6 / xhigh  | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 5.6 / max    | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 5.6 / ultra  | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6 / low      | 100.00    | 94.64         | 100.00         | 100.00         |
| Sol 6 / medium   | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6 / high     | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6 / xhigh    | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6 / max      | 100.00    | 100.00        | 100.00         | N/A            |
| Sol 6 / ultra    | N/A       | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / low    | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / medium | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / high   | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / xhigh  | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / max    | 100.00    | 100.00        | 100.00         | 100.00         |
| Sol 6.1 / ultra  | 100.00    | 100.00        | 100.00         | 100.00         |

**Boundary defects:** Sol 5.6 / Low r2 and Sol 6 / Low r1 cache constructors fail on a finite integer TTL of  $10^{**}400$  with an integer clock. Applying `math.isfinite` to that integer raises `OverflowError`. Their original acceptance tests passed.

Expanded tests add 1,200 random unions, cache model traces, DAG failure graphs and concurrency barriers, plus reservation stock-model steps, races and persistence/error cases. Properties/boundaries/concurrency weigh 40/30/30, renormalized when a task lacks a category.

N/A is incomplete completed-code coverage. The provider-failed stub and saved timeout code remain in raw evidence; they are excluded from completed-code means. The new tasks use their own frozen behavioral rubric, without retroactively changing old scores.

# Do the added tests detect defects?

Per-task mutation sensitivity, averaged across two completed candidates. N/A shows usable-score coverage for that pair.

| Model / effort   | Intervals | Cache     | DAG       | Service   | Optimizer | MVCC      |
|------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Sol 5.6 / low    | 50%       | 50%       | 70%       | 58%       | 0%        | 0%        |
| Sol 5.6 / medium | 100%      | 60%       | 70%       | 0%        | 0%        | 50%       |
| Sol 5.6 / high   | 100%      | 60%       | 90%       | 0%        | N/A (1/2) | 42%       |
| Sol 5.6 / xhigh  | 50%       | 60%       | 80%       | 0%        | 90%       | 42%       |
| Sol 5.6 / max    | 50%       | 60%       | 100%      | 0%        | 90%       | 67%       |
| Sol 5.6 / ultra  | 50%       | 60%       | N/A (1/2) | 0%        | N/A (1/2) | 58%       |
| Sol 6 / low      | 0%        | 30%       | 70%       | 0%        | 40%       | 25%       |
| Sol 6 / medium   | 88%       | 60%       | 100%      | N/A (1/2) | N/A (0/2) | 75%       |
| Sol 6 / high     | 88%       | 60%       | 100%      | N/A (0/2) | 100%      | 67%       |
| Sol 6 / xhigh    | 100%      | N/A (0/2) | 100%      | N/A (0/2) | 90%       | 83%       |
| Sol 6 / max      | 100%      | N/A (1/2) | 100%      | N/A (0/2) | 90%       | N/A (1/2) |
| Sol 6 / ultra    | N/A (1/2) | 70%       | 100%      | N/A (0/2) | N/A (1/2) | 83%       |
| Sol 6.1 / low    | 100%      | 70%       | 80%       | 100%      | N/A (1/2) | 75%       |
| Sol 6.1 / medium | 100%      | 80%       | 100%      | N/A (0/2) | N/A (0/2) | 92%       |
| Sol 6.1 / high   | 100%      | 80%       | 100%      | N/A (0/2) | N/A (0/2) | 100%      |
| Sol 6.1 / xhigh  | 100%      | 80%       | 100%      | N/A (0/2) | N/A (0/2) | 100%      |
| Sol 6.1 / max    | 100%      | 80%       | 100%      | N/A (0/2) | N/A (0/2) | 100%      |
| Sol 6.1 / ultra  | 100%      | 90%       | 100%      | N/A (0/2) | N/A (0/2) | 100%      |

175/214 completed candidates have usable scores; 29 have no recognized added tests. On the complex tasks, 55/72 scores are usable and 10 candidates have no recognized added tests.

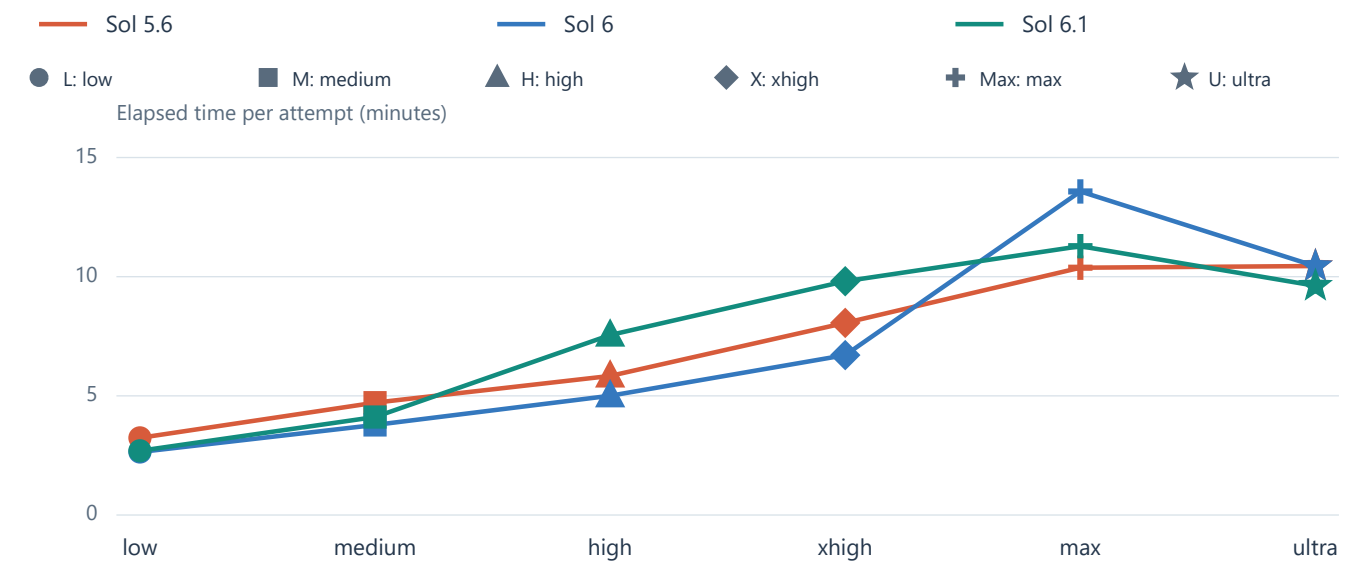
Discovery uses unittest-compatible tests/test\*.py, excluding supplied test\_public.py. Tests must pass on both candidate and positive control. No added tests score zero only for this criterion. Import errors, skips, timeouts and reference incompatibility are inconclusive.

Optimizer test strength is unavailable for 16/36 candidates: twelve positive-control test runs hit the 30-second timeout, three encounter import/execution issues, and one mutant run is inconclusive. These evaluator limitations particularly affect Sol 6.1 optimizer coverage and do not establish a candidate-code defect.

Fixed sets: 4 interval, 5 cache, 5 DAG, 6 service, 5 optimizer and 6 MVCC defects. The original four-task classifier detects an executed test failure or error after the reference passed; loader failures/timeouts are inconclusive. The complex-task classifier counts assertion failures only and treats execution errors/hangs as unavailable. Interpret sensitivity within each task.

# Agent time: a secondary comparison

Time remains useful for workflow planning, separate from the primary money-versus-quality comparison. The top curve shows mean elapsed agent minutes; the table reports each task in seconds.



| Model / effort   | Intervals | Cache | DAG | Service | Optimizer | MVCC |
|------------------|-----------|-------|-----|---------|-----------|------|
| Sol 5.6 / low    | 69        | 108   | 125 | 280     | 350       | 233  |
| Sol 5.6 / medium | 91        | 130   | 165 | 399     | 621       | 290  |
| Sol 5.6 / high   | 108       | 148   | 198 | 531     | 755       | 360  |
| Sol 5.6 / xhigh  | 147       | 141   | 215 | 692     | 1,298     | 410  |
| Sol 5.6 / max    | 179       | 212   | 305 | 908     | 1,584     | 548  |
| Sol 5.6 / ultra  | 151       | 225   | 314 | 941     | 1,524     | 606  |
| Sol 6 / low      | 68        | 78    | 110 | 253     | 240       | 204  |
| Sol 6 / medium   | 88        | 112   | 145 | 299     | 490       | 226  |
| Sol 6 / high     | 77        | 119   | 168 | 505     | 683       | 249  |
| Sol 6 / xhigh    | 194       | 242   | 209 | 666     | 762       | 343  |
| Sol 6 / max      | 219       | 275   | 301 | 2,380   | 1,220     | 492  |
| Sol 6 / ultra    | 158       | 215   | 215 | 1,230   | 1,493     | 447  |
| Sol 6.1 / low    | 72        | 96    | 123 | 272     | 221       | 184  |
| Sol 6.1 / medium | 81        | 105   | 152 | 413     | 490       | 240  |
| Sol 6.1 / high   | 165       | 166   | 268 | 871     | 863       | 386  |
| Sol 6.1 / xhigh  | 121       | 158   | 325 | 1,028   | 1,394     | 506  |
| Sol 6.1 / max    | 230       | 224   | 385 | 1,096   | 1,580     | 545  |
| Sol 6.1 / ultra  | 134       | 176   | 420 | 1,068   | 1,198     | 466  |

Elapsed agent time includes reasoning, commands, self-tests and waits during the turn. Setup, snapshotting and external grading are excluded. Task limits are 10 / 20 / 40 / 40 / 40 / 40 minutes; interrupted durations remain included.

# Generated-code runtime and allocation

Existing quality-v2.1 runtime / Python-allocation measurements for the first four tasks. Each cell is milliseconds / KiB.

| Model / effort   | Intervals     | TTL/LRU cache | Concurrent DAG | SQLite service |
|------------------|---------------|---------------|----------------|----------------|
| Sol 5.6 / low    | 17.90 / 4,336 | 34.68 / 104   | 54.16 / 953    | 515.90 / 4     |
| Sol 5.6 / medium | 16.34 / 4,454 | 34.95 / 104   | 61.10 / 901    | 478.92 / 4     |
| Sol 5.6 / high   | 18.84 / 4,454 | 35.75 / 104   | 63.61 / 870    | 533.39 / 4     |
| Sol 5.6 / xhigh  | 17.02 / 4,454 | 35.53 / 104   | 60.55 / 877    | 553.25 / 5     |
| Sol 5.6 / max    | 16.83 / 4,336 | 34.36 / 104   | 56.91 / 755    | 648.41 / 5     |
| Sol 5.6 / ultra  | 17.61 / 4,336 | 33.78 / 104   | 53.81 / 863    | 611.56 / 4     |
| Sol 6 / low      | 13.95 / 4,336 | 39.05 / 112   | 54.40 / 802    | 537.83 / 4     |
| Sol 6 / medium   | 15.54 / 4,336 | 19.90 / 112   | 58.85 / 806    | 543.75 / 4     |
| Sol 6 / high     | 15.16 / 4,336 | 40.23 / 112   | 55.15 / 730    | 693.05 / 4     |
| Sol 6 / xhigh    | 16.97 / 4,336 | 40.44 / 112   | 51.28 / 813    | 622.31 / 4     |
| Sol 6 / max      | 16.25 / 4,336 | 38.47 / 112   | 57.80 / 844    | N/A            |
| Sol 6 / ultra    | N/A           | 36.34 / 108   | 57.56 / 838    | 504.32 / 4     |
| Sol 6.1 / low    | 17.07 / 4,336 | 33.74 / 104   | 77.32 / 781    | 529.36 / 5     |
| Sol 6.1 / medium | 17.29 / 4,336 | 34.85 / 104   | 71.42 / 834    | 861.48 / 5     |
| Sol 6.1 / high   | 16.68 / 4,336 | 35.37 / 104   | 66.49 / 806    | 555.97 / 5     |
| Sol 6.1 / xhigh  | 16.82 / 4,336 | 35.33 / 104   | 68.93 / 730    | 468.35 / 5     |
| Sol 6.1 / max    | 17.24 / 4,336 | 38.25 / 104   | 55.46 / 704    | 511.43 / 5     |
| Sol 6.1 / ultra  | 17.30 / 4,336 | 34.98 / 104   | 52.31 / 705    | 467.98 / 5     |

One warm-up, then the median of three sequential runs. Workloads: 30,000 intervals; cache capacity 1,000 with 1,000 writes and 3,000 reads; 1,500 DAG nodes; 40 reservation cycles. Allocation is a separate tracemalloc measurement, not process RSS or native/SQLite memory. These values describe execution after generation, separate from agent completion time. The complex tasks have their scored scale/trace checks; equivalent runtime and memory profiles were not collected for them. Small differences can reflect scheduling noise.

# Code quality and design review

Behavior, test strength and execution efficiency are reported separately. Manual rubric scoring remains pending.

| Evidence layer        | Scope            | Interpretation                                                      |
|-----------------------|------------------|---------------------------------------------------------------------|
| Main functional score | All six tasks    | Equal task weights; fixed category weights for the complex tasks.   |
| Expanded correctness  | First four tasks | Frozen properties, boundaries and concurrency checks.               |
| Own-test strength     | All six tasks    | Sensitivity to fixed per-task contract mutations.                   |
| Runtime / memory      | First four tasks | Descriptive generated-code measurements, unscored.                  |
| Manual design rubric  | Pending          | Clarity, design and failure maintenance, with source/line evidence. |

## Manual rubric

| Area                | Evidence a reviewer should inspect                                    |
|---------------------|-----------------------------------------------------------------------|
| Clarity             | Intent, names, coherent control flow, explanations and useful tests.  |
| Design              | Interfaces, responsibilities, cohesion, duplication and contract fit. |
| Failure maintenance | Error handling, cleanup, transactions, concurrency and safe changes.  |

The existing four-task quality formula uses 30% original correctness, 35% expanded correctness, 20% own-test strength and 15% manual review. Its normalized 85-point automatic evidence is not a full quality score. It is preserved for those tasks and is not extrapolated to the added tasks.

Static line/function counts are review signals, not grades. Tests tied to private implementation details can be valid for the candidate while yielding inconclusive cross-implementation mutation evidence.

# Method, token prices and reproducibility

Each attempt uses one turn, a fresh session/workspace, the exact requested model/effort and Standard speed. Task prompt hashes are identical within each task across all 36 attempts. SHA-256 manifests protect the prepared inputs and saved candidate code.

| Model   | Input  | Cached input | Cache write | Output  |
|---------|--------|--------------|-------------|---------|
| Sol 5.6 | \$4.00 | \$0.40       | \$5.00      | \$20.00 |
| Sol 6   | \$2.00 | \$0.20       | \$2.50      | \$10.00 |
| Sol 6.1 | \$2.00 | \$0.10       | \$2.50      | \$10.00 |

Rates are USD per one million tokens, frozen on 30 Sep 2026 and verified on 1 Oct. The same schedule is used throughout for comparability. This report does not assert a newly checked current price. Ultra is a verified Codex effort setting; the USD calculation does not imply an API Ultra option.

Uncached input, cached reads, cache writes and output are priced without double counting. Reasoning tokens are part of output. Per-request input above 272,000 tokens applies the frozen 2x input/cache and 1.5x output multipliers. Hosted-tool fees and regional premiums are excluded.

Exact token estimates: 214/216. Observed total is at least \$87.59; two partial-usage attempts prevent a complete total. The two complex tasks have exact observed token cost \$41.81 across 72 attempts.

Secondary token-based credit estimate: 2,189.95 observed credits, including partial usage. Shared account balance changes do not identify billed credits for an individual run.

Pinned runtime: codex-cli 0.159.0; Node v22.18.0; Python 3.14.0 on Windows. Shell/tool recovery is included in agent time. Read isolation is instruction-based; automated command scans are diagnostics.

## Sources and evidence

Sol 5.6 API rates | Sol 6 API rates | Sol 6.1 API rates  
Codex credits | Codex app-server

Public evidence: results/results.json, quality-v2.1.json, complex-test-strength.json, report-audit.json and provenance.json; candidates/ and per-attempt prompt/final evidence. The original local audit verified all 216 snapshots/settings/turns and price calculations. Raw RPC/session logs are not distributed. Six insight views and results/insights.json derive from the same measurements, with no new inference or rescoring.

Public results SHA-256: b20d615c1e4334911db470e27fc163f31efa3a14e32161cf3a57e2aea0ab5fbe  
Original source identities and both prior reports are recorded in results/provenance.json. Current insight presentation: 5 Oct 2026.